

Single-Image Internal Geometry Generation with TRELLIS

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Abstract

TRELLIS generates high-quality 3D assets from single images, but struggles to reconstruct internal structures. This limitation stems from two main factors: the original training data, primarily derived from Objaverse, contains little meaningful internal geometry, and the rendering-based reconstruction objective used to supervise the VAE decoder provides no direct supervision for internal geometry. We address these limitations with two complementary modifications. First, we apply LoRA fine-tuning to the VAE decoder and both flow-matching networks on ShapeNet objects with internal structures. For the VAE decoder, we additionally introduce a cross-section rendering loss to provide direct supervision for internal geometry. Second, we introduce a retrieval-based refinement pipeline. It retrieves ShapeNet training objects with similar exterior geometry, aligns them to the generated object, and transfers compatible interior components. A learned confidence model accepts a retrieved modification only when it is predicted to improve the reconstruction. Experiments on ShapeNet show that interior-aware fine-tuning substantially improves internal geometry generation, while retrieval-based refinement further enhances interior quality with finer and more realistic structures.

1. Introduction

Recent advances in 3D generative models enable high-quality 3D reconstruction from a single image. TRELLIS [4] achieves strong results across diverse object categories, but often fails to generate meaningful internal structures. This limits applications that require complete object geometry rather than only an accurate exterior surface.

Generating internal geometry from a single image is challenging for three main reasons. First, TRELLIS is trained primarily on data derived from Objaverse, where objects rarely contain meaningful internal structures. As a result, the model has limited examples from which to learn common interior layouts. Second, the rendering-based reconstruction objective used to supervise the Variational Au-

toencoder (VAE) decoder relies only on exterior views, providing no direct supervision for internal geometry. Third, the task is inherently ambiguous: the input image contains no direct observation of the object’s interior, and multiple internal geometries may be consistent with the same visible exterior.

We address these limitations with two complementary approaches. First, we apply Low-Rank Adaptation (LoRA) fine-tuning to the sparse-structure flow model, the Structured Latent (SLAT) flow model, and the mesh VAE decoder using ShapeNet objects with internal structures. For the VAE decoder, we additionally introduce cross-section rendering supervision, which exposes internal geometry during training and provides a direct reconstruction signal. Second, we introduce a retrieval-based refinement pipeline. Given a prediction from the interior-aware fine-tuned TRELLIS model, we retrieve training shapes with similar exterior geometry, align them to the generated object, and transfer compatible interior components while preserving the predicted exterior. A learned confidence model accepts a retrieved modification only when it is predicted to improve the internal reconstruction.

Our main contributions are:

- We adapt TRELLIS to generate internal geometry by applying LoRA fine-tuning to its sparse structure flow model, SLAT flow model, and mesh VAE decoder on ShapeNet objects with internal structures, together with additional cross-section rendering supervision for the VAE decoder.
- We introduce a retrieval-based refinement pipeline that retrieves training shapes with similar exterior geometry, aligns them to TRELLIS predictions, and transfers compatible interior components while preserving the generated exterior.
- We develop a learned confidence model that accepts retrieved modifications only when they are predicted to improve internal reconstruction quality.
- We provide comprehensive quantitative and qualitative evaluations demonstrating the effectiveness of both interior-aware fine-tuning and retrieval-based refinement for TRELLIS on the ShapeNet dataset.

2. Related Work

Image-conditioned 3D generation. TRELLIS [4] is a large-scale 3D generative model based on a structured latent representation (SLAT) learned by a transformer-based VAE. Image-conditioned generation proceeds through two flow-matching stages. A sparse-structure flow model first generates a coarse occupancy structure in the latent space of a separate 3D-CNN-based Sparse Structure VAE. A SLAT flow model then generates local latent features on the resulting active structure, which are decoded into the final 3D asset. Our work builds directly on TRELLIS by adapting its pretrained pipeline for interior-aware generation and introducing a retrieval-based refinement stage.

Shape datasets. ShapeNet [1] is a large-scale dataset of category-annotated 3D CAD models. Unlike Objaverse, whose objects often contain little meaningful internal geometry, several ShapeNet categories include explicit internal structures such as vehicle interiors, shelves, and compartments. We therefore use ShapeNet both for interior-aware fine-tuning and as the donor database for retrieval-based refinement.

Low-Rank Adaptation. LoRA [2] enables parameter-efficient fine-tuning by keeping pretrained weights fixed and learning low-rank updates to selected weight matrices. This substantially reduces the number of trainable parameters while preserving the knowledge of the pretrained model. We use LoRA to fine-tune the TRELLIS sparse structure flow model, SLAT flow model, and mesh VAE decoder on ShapeNet objects with internal structures.

Visual feature extraction. DINOv2 [3] learns general-purpose visual features through self-supervised pretraining. It is the default feature extractor used in TRELLIS for representation learning and image-conditioned generation. In our retrieval stage, we additionally use DINOv2 features to identify visually similar ShapeNet objects as candidate donors for interior refinement.

3. Method

3.1. Fine-Tuning for Internal Geometry Generation

We fine-tune the sparse structure flow model, the SLAT flow model, and the mesh VAE decoder of the pretrained TRELLIS pipeline on our ShapeNet dataset with internal geometry. For each component, LoRA adapters are inserted into every linear layer.

Mesh VAE Decoder. The mesh VAE decoder is fine-tuned with LoRA using a modified reconstruction objective designed to provide direct supervision for internal geometry. As illustrated in Fig. 1, the mesh reconstructed from the ground-truth SLAT latents is supervised through two rendering branches: the original exterior rendering loss and an additional cross-section rendering loss.

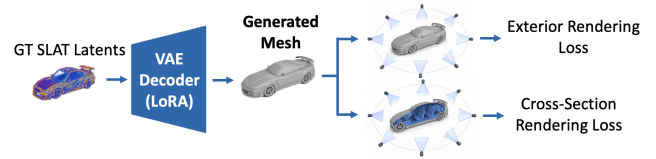


Figure 1. LoRA fine-tuning of the mesh VAE decoder during training. The decoder reconstructs meshes from ground-truth SLAT latents and is supervised by the original exterior rendering loss together with an additional cross-section rendering loss.

In the original TRELLIS objective, the reconstructed mesh is rendered from a set of exterior camera views and compared with corresponding renderings of the ground-truth mesh. While this provides strong supervision for visible surfaces, internal geometry is not directly observed by these views and therefore receives no direct reconstruction signal. To expose this hidden geometry, we additionally slice both the ground-truth mesh and its reconstruction with a plane passing through the object center and perpendicular to the camera’s principal axis. The resulting cross-sectional views are rendered and compared using the same rendering-based supervision as the exterior views. The modified reconstruction objective is:

$$\mathcal{L}'_{\text{render}} = \mathcal{L}_{\text{render}}(M, \hat{M}) + \lambda_c \mathcal{L}_{\text{render}}(S(M), S(\hat{M})) \quad (1)$$

where M and $\hat{M}$ denote the ground-truth and reconstructed meshes, respectively, $S(\cdot)$ denotes the slicing operation, and λ_c controls the contribution of the cross-section term. This additional supervision enables internal geometry to contribute directly to the decoder reconstruction objective.

We verify that the fine-tuned decoder can decode ground-truth SLATs to objects with interior structures before proceeding with fine-tuning the flow-matching models.

Sparse-Structure Flow Model. Unlike SLAT features, which are explicitly associated with occupied spatial locations, the sparse-structure flow operates in the compressed latent space of the 3D-CNN-based Sparse Structure VAE, where there is no direct correspondence between individual latent features and voxels classified as internal or external. Consequently, we decided to fine-tune the sparse structure flow model using LoRA with its original flow-matching objective unchanged.

SLAT Flow Model. For the SLAT flow model, we modify the original flow-matching objective to place greater emphasis on latent features associated with internal voxels. As illustrated in Fig. 2, voxels belonging to the set of internal structure voxels, Ω_{int} , receive an increased loss weight λ_{int} , while all remaining voxels are assigned a weight of 1.

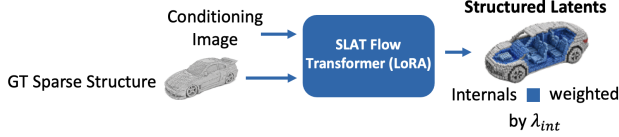


Figure 2. LoRA fine-tuning of the SLAT flow model. During training, the SLAT flow model is conditioned on the input image and the ground-truth sparse structure. Internal latent voxels (blue) receive an increased loss weight λ_{int} , while all remaining voxels are assigned a weight of 1.

The resulting objective is:

$$\mathcal{L}_{\text{SLAT}} = \frac{1}{N} \sum_{i=1}^N w_i \|\hat{\mathbf{v}}_i - \mathbf{v}_i\|_2^2, \quad (2)$$

$$w_i = \begin{cases} \lambda_{\text{int}}, & i \in \Omega_{\text{int}}, \\ 1, & \text{otherwise.} \end{cases}$$

Here, $\hat{\mathbf{v}}_i$ and $\mathbf{v}_i$ denote the predicted and target flow velocities at voxel i , respectively. We identify Ω_{int} using a simple proxy. For each coordinate axis, occupied voxels are considered along lines parallel to that axis, and the two outermost occupied voxel layers on either side of the object are marked as exterior. This weighting increases the contribution of internal regions to the flow-matching objective, directing more of the fine-tuning signal toward internal geometry.

3.2. Retrieval-Based Interior Refinement

The retrieval pipeline treats the fine-tuned TRELIS output as a base reconstruction and uses complete training shapes with similar appearance as priors for refining its internal geometry. The generated exterior is preserved throughout the pipeline, and retrieval is only applied when a proposed modification is predicted to improve the base reconstruction. An overview of the retrieval pipeline is shown in Fig. 3.

Retrieval and alignment. We voxelize the generated mesh at a resolution of 64^3 . Using the boundary-based classification described in Section 3.1, we identify the exterior, which defines a protected volume into which retrieved geometry

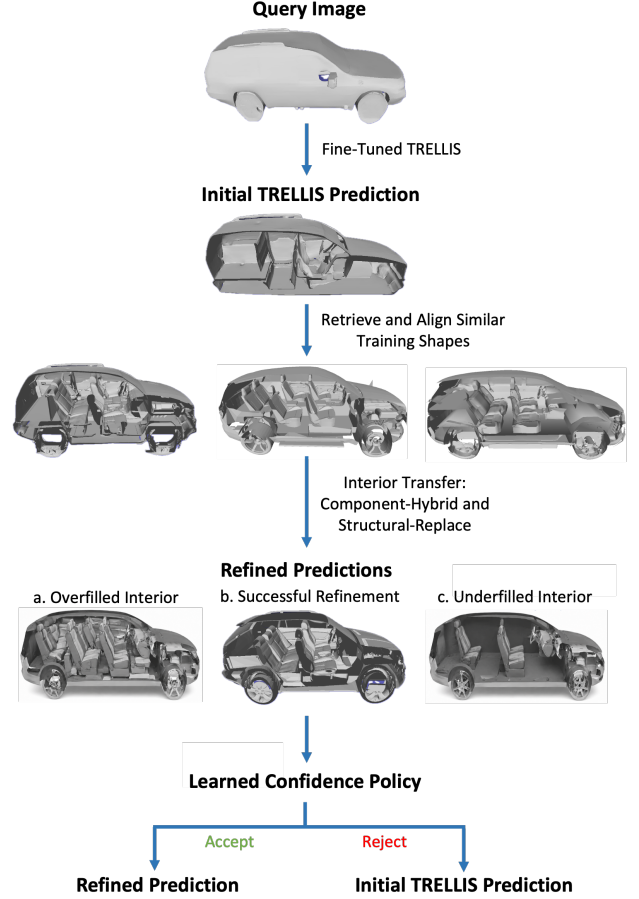


Figure 3. Overview of the retrieval-based interior refinement pipeline. Similar training shapes are retrieved and aligned to the initial fine-tuned TRELIS prediction, producing candidate repairs through component-hybrid and structural-replace interior transfer. A learned confidence policy selects an improved candidate or retains the original prediction.

may be inserted.

For each input, we retrieve the 20 most similar training shapes from the same ShapeNet category using DINOv2

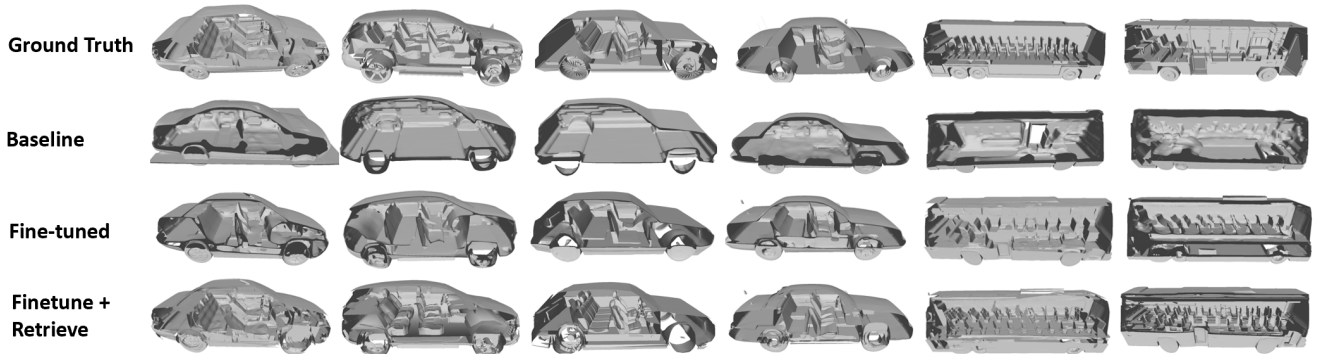


Figure 4. Qualitative comparison of original TRELIS, interior-aware fine-tuning, and retrieval-based refinement.

features of the conditioning images. Each donor is aligned to the generated exterior using an axis-aligned affine transformation and ranked according to appearance similarity, exterior agreement, interior compatibility, donor consensus, and compatibility with the protected volume. The eight highest-ranked donors are retained.

Interior repair. For each shortlisted donor, we decompose the aligned interior into connected components and discard components that violate constraints on size, connectivity, containment within the protected volume, or donor support. Each donor produces two repair variants. The *component-hybrid* variant inserts valid donor components while retaining non-conflicting TRELIS geometry, which may lead to *overfilled* interiors (Fig. 3). The *structural-replace* variant replaces the generated interior within the protected volume with donor components, which may produce *underfilled* interiors (Fig. 3). In total, eight donors yield 16 repair candidates.

Candidate selection and mesh reconstruction. We train a kNN regressor to predict the change in interior F1 of each candidate relative to the fine-tuned TRELIS reconstruction. Training targets are obtained by evaluating candidates against their ground-truth interiors. The regressor uses only inference-time features, including image similarity, exterior agreement, donor support, repair type, transferred and removed geometry, and interior statistics.

At inference, the highest-scoring candidate is accepted only if its predicted improvement exceeds a confidence threshold; otherwise, the original fine-tuned TRELIS prediction is retained. For the final mesh, affected interior triangles are replaced by the corresponding triangles from the aligned donor, preserving the TRELIS exterior while transferring the donor’s detailed interior geometry.

4. Experiments

We evaluate our approach on four ShapeNet categories containing meaningful internal structures: *car*, *bus*, *cabinet*, and *file cabinet*. Our dataset comprises 1,560 objects for fine-tuning and 195 held-out objects for evaluation. LoRA is applied on the mesh decoder, sparse structure flow and SLAT flow model with ranks 32, 16 and 8 respectively. We set the strength of the cross section rendering loss to $\lambda_c = 1.0$ and the weight for internal structure voxels to $w_i = 5.0$ empirically.

4.1. Quantitative Results

We evaluate the reconstruction of internal geometry using Precision, Recall, F1 score, and Chamfer Distance (CD). Precision, Recall, and F1 measure agreement in interior occupancy, while CD measures the geometric distance between predicted and ground-truth interior surfaces.

We compare the original TRELIS model, our interior-aware fine-tuned model, and the final retrieval-refined out-

Table 1. Quantitative evaluation of internal geometry on the ShapeNet test set.

Method	Precision $\uparrow$	Recall $\uparrow$	F1 $\uparrow$	CD $\downarrow$
TRELIS	0.198	0.081	0.102	0.075
Fine-tuned	0.246	0.277	0.242	0.043
Retrieval-refined	0.319	0.352	0.256	0.044

put on the 195 held-out ShapeNet objects. The quantitative results are summarized in Table 1.

Interior-aware fine-tuning substantially improves all occupancy-based metrics and reduces CD, demonstrating that TRELIS can learn to generate meaningful internal geometry when trained with appropriate supervision. Retrieval-based refinement further improves Precision, Recall, and F1 by transferring high-quality interior structures from geometrically compatible training shapes, while maintaining a comparable CD.

4.2. Qualitative Results

Representative reconstructions are shown in Fig. 4. Compared with the original TRELIS model, interior-aware fine-tuning enables the generation of plausible internal structures. However, the generated interiors remain relatively generic and often lack fine geometric detail, likely due to the limited size and diversity of the fine-tuning dataset.

Retrieval-based refinement complements the learned model by transferring high-detail interior structures from similar training shapes while preserving the generated exterior, resulting in more complete and realistic internal geometry.

5. Conclusion & Outlook

We investigated internal geometry generation in TRELIS, which in its original form produces little meaningful structure inside reconstructed objects.

Interior-aware fine-tuning with cross-section supervision enables TRELIS to generate plausible interiors from ShapeNet objects with internal structure, although the results remain generic and lack fine geometric detail.

Retrieval-based refinement further improves interior fidelity by transferring compatible geometry from similar training shapes. Together, these results show that fine-tuning and retrieval provide an effective framework for single-image internal geometry generation.

Future work could improve the learned prior using a larger and more diverse dataset with detailed internal geometry. Another direction is finer-grained semantic conditioning through cross-attention to subtype tokens, providing more specific priors for internal structure.

References

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